Experiment-1

Network Commands: Running and using services/commands like ping, traceroute, nslookup, arp, telnet, ftp, etc.

Using ipconfig command

ipconfig (standing for "Internet Protocol configuration") is a console application program of some computer operating systems that displays all current TCP/IP network configuration values and refreshes Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) settings.

```
Microsoft Windows [Version 10.0.15063]
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C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : kiet.edu
   Link-local IPv6 Address . . . . . : fe80::acc0:7223:8621:3e75%7
   IPv4 Address. . . . . . . . . . . : 10.21.73.96
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.21.73.1
```

Using ping command

Ping is a computer network administration software utility used to test the reachability of a host on an Internet Protocol network. It is available for virtually all operating systems that have networking capability, including most embedded network administration software.

```
C:\Windows\system32>ping 10.21.73.32

Pinging 10.21.73.32 with 32 bytes of data:
Reply from 10.21.73.32: bytes=32 time<1ms TTL=128
Reply from 10.21.73.32: bytes=32 time<1ms TTL=128
Reply from 10.21.73.32: bytes=32 time<1ms TTL=128
Reply from 10.21.73.32: bytes=32 time=2ms TTL=128

Ping statistics for 10.21.73.32:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Using tracert command

The tracert command is a Command Prompt command that's used to show several details about the path that a packet takes from the computer or device you're on to whatever destination you specify.

Windows: **tracert** [**-d**] [**-h** *MaxHops*] [**-w** *TimeOut*] [**-4**] [**-6**] *target* [**/?**]

Linux: **traceroute** [**-d**] [**-h** *MaxHops*] [**-w** *TimeOut*] [**-4**] [**-6**] *target* [**/?**]

```
C:\Users\email>tracert facebook.com

Tracing route to facebook.com [2a03:2880:f1ff:83:face:b00c:0:25de]
over a maximum of 30 hops:

  1    41 ms    36 ms    37 ms  2401:4900:40f1:b636:0:4a:3a22:9c01
  2    68 ms    57 ms    47 ms  2401:4900:40f1:b636:0:4a:3a22:9c40
  3     *        *        *     Request timed out.
  4    50 ms    42 ms    35 ms  2401:4900:0:c000::1:b1
  5    46 ms    34 ms    51 ms  2401:4900:0:c000::1:d2
  6    67 ms    46 ms    68 ms  2404:a800:1a00:803::69
  7   119 ms    34 ms    41 ms  ae20.pr04.del1.tfbnw.net [2620:0:1cff:dead:beee::bac]
  8    51 ms    55 ms    38 ms  po104.psw04.del1.tfbnw.net [2620:0:1cff:dead:bef0::471]
  9    65 ms    40 ms    35 ms  po4.msw1ar.01.del1.tfbnw.net [2a03:2880:f044:ffff::bf]
 10    84 ms    49 ms    38 ms  edge-star-mini6-shv-01-any2.facebook.com [2a03:2880:f1ff:83:face:b00c:0:25de]

Trace complete.
```

Using hostname command

Use this command to display the IP address of the remote machine.

hostname [-v] [-a] [--alias] [-d] [--domain] [-f] [--fqdn] [-A] [--all-fqdns] [-i] [--ip-address] [-I] [--all-ip-addresses] [--long] [-s] [--short] [-y] [--yp] [--nis]

```
C:\Windows\system32>hostname
CSLAB-2-27
```

Using netstat command

In computing, netstat is a command-line network utility that displays network connections for Transmission Control Protocol, routing tables, and a number of network interface and network protocol statistics.

netstat [address_family_options] [--tcp|-t] [--udp|-u] [--raw|-w] [--listening|-l] [--all|-a] [--numeric|-n] [--numeric-hosts][--numeric-ports][--numeric-ports] [--symbolic|-N] [--extend|-e[--extend|-e]] [--timers|-o] [--program|-p] [--verbose|-v] [--continuous|-c] [delay]

```
C:\Windows\system32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.21.73.96:49672      40.90.189.152:https    ESTABLISHED
  TCP    10.21.73.96:49760      ec2-34-214-115-165:https  ESTABLISHED
  TCP    10.21.73.96:49824      dump:microsoft-ds      ESTABLISHED
  TCP    10.21.73.96:49900      maa05s12-in-f4:https   TIME_WAIT
  TCP    10.21.73.96:49905      bom07s29-in-f3:https   TIME_WAIT
  TCP    10.21.73.96:49909      del12s05-in-f14:https  TIME_WAIT
  TCP    10.21.73.96:49922      maa03s31-in-f3:https   TIME_WAIT
  TCP    10.21.73.96:49924      del03s15-in-f14:https  TIME_WAIT
  TCP    10.21.73.96:49925      a-0001:https           CLOSE_WAIT
  TCP    10.21.73.96:49926      13.107.9.158:https     CLOSE_WAIT
  TCP    10.21.73.96:49927      13.107.19.254:https    CLOSE_WAIT
  TCP    10.21.73.96:49929      13.107.9.254:https     CLOSE_WAIT
  TCP    10.21.73.96:49930      51.140.152.167:https   CLOSE_WAIT
  TCP    10.21.73.96:49931      117.18.237.29:http     CLOSE_WAIT
  TCP    10.21.73.96:49932      204.79.197.222:https   CLOSE_WAIT
  TCP    10.21.73.96:49933      52.109.124.116:https   TIME_WAIT
  TCP    10.21.73.96:49934      52.109.56.46:https     TIME_WAIT
  TCP    10.21.73.96:49935      a23-63-110-112:https   TIME_WAIT
  TCP    10.21.73.96:49936      182.19.89.107:https    TIME_WAIT
```

Using arp command

arp command manipulates the System's ARP cache. It also allows a complete dump of the ARP cache. ARP stands for Address Resolution Protocol. The primary function of this protocol is to resolve the IP address of a system to its MAC address, and hence it works between level 2(Data link layer) and level 3(Network layer).

```
C:\Windows\system32>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.
```

```
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.
```

```
C:\Users\email>arp -a

Interface: 192.168.43.183 --- 0x16
  Internet Address      Physical Address      Type
  192.168.43.1          00-27-15-62-66-69     dynamic
  192.168.43.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Using nslookup command

nslookup is a network administration command-line tool for querying the Domain Name System to obtain the mapping between domain name and IP address, or other DNS records.

nslookup [HOST] [SERVER]

```
C:\Users\email>nslookup facebook.com
Server:  UnKnown
Address:  2401:4900:50:9::18e

Non-authoritative answer:
Name:    facebook.com
Addresses:  2a03:2880:f1ff:83:face:b00c:0:25de
          69.171.250.35
```

Experiment-2

Aim of the experiment: Making of cross cable and straight cable.

Apparatus/Tools/Equipments/Components Required:

1. RJ-45 connector,
2. Crimping Tool,
3. Twisted pair Cable,
4. Cable Tester.

Procedure:

1. Start by stripping off about 2 inches of the plastic jacket off the end of the cable. Be very careful at this point, as to not nick or cut into the wires, which are inside. Doing so could alter the characteristics of your cable, or even worse render is useless. Check the wires, one more time for nicks or cuts. If there are any, just whack the whole end off, and start over.
2. Spread the wires apart, but be sure to hold onto the base of the jacket with your other hand. You do not want the wires to become untwisted down inside the jacket. Category 5 cable must only have 1/2 of an inch of 'untwisted' wire at the end; otherwise it will be 'out of spec'. At this point, you obviously have ALOT more than 1/2 of an inch of un-twisted wire.
3. You have 2 end jacks, which must be installed on your cable. If you are using a pre-made cable, with one of the ends whacked off, you only have one end to install - the crossed over end. Below are two diagrams, which show how you need to arrange the cables for each type of cable end. Decide at this point which end you are making and examine the associated picture below.

Diagram shows you how to prepare Cross wired connection

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Diagram shows you how to prepare straight through wired connection

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Cable Crimping Steps:

1. Remove the outmost vinyl shield for 12mm at one end of the cable (we call this side A-side).
2. Arrange the metal wires in parallel
3. Insert the metal wires into RJ45 connector on keeping the metal wire arrangement.
4. Set the RJ45 connector (with the cable) on the pliers, and squeeze it tightly.
5. Make the other side of the cable (we call this side B-side) in the same way.


Testing the crimped cable using a cable tester:

Step 1: Skin off the cable jacket 3.0 cm long cable stripper up to cable
Step 2: Untwist each pair and straighten each wire 190 0 1.5 cm long.
Step 3: Cut all the wires.

Step 4: Insert the wires into the RJ45 connector right white orange left brown the pins facing up
Step 5: Place the connector into a crimping tool, and squeeze hard so that the handle reaches its full swing.
Step 6: Use a cable tester to test for proper continuity.

Result:

Cable Crimping, straight Cabling and Cross Cabling, and testing the crimped cable using a cable tester are done successfully.